Brac University

Department of Electrical & Electronic Engineering

Semester Spring-25

Course Number: EEE103L

Course Title: Computer Programming Laboratory

Section: 06

Lab Report

Experiment no. 04

Name of the experiment: Familiarization with Functions and Recursion

Prepared by:

Name: **Tanzeel Ahmed** ID: **24321367**

Group Number: 04

Other Group members:

Sl.	***ID***	***Name***
1.	24121083	Abontika Das
2.	24121181	Tasnim Sultana
3.	24121249	Faria Rahman Oyshi
4.	24121251	Yeasin Khan Sifat

Lab report of experiment 4

a. Write a C program to find whether a number is Palindrome or Not using functions.

Code:

```
#include <stdio.h>

int reverse_num(int num)

{

int reverse = 0;

while (num > 0)

{

reverse = reverse * 10 + num % 10;

num = num/10;

}

return reverse;

}

int palindrome(int num)

{

return num == reverse_num(num);

}

int main()

{

int number;

printf("Enter a number: ");

scanf("%d", &number);

if(palindrome(number))

printf("%d is a Palindrome number.\n", number);
```

else

printf("%d is Not a Palindrome number.\n", number);

return 0;

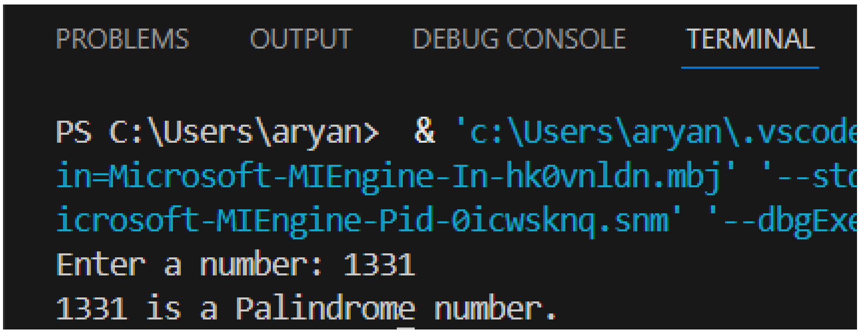
}

<u>Screenshot of code:</u>

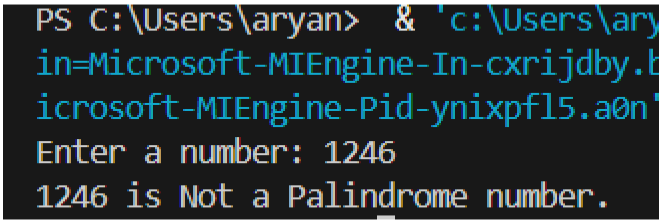

```c
#include <stdio.h>

int reverse_num(int num)
{
    int reverse = 0;
    while (num > 0)
    {
        reverse = reverse * 10 + num % 10;
        num = num/10;
    }
    return reverse;
}

int palindrome(int num)
{
    return num == reverse_num(num);
}

int main()
{
    int number;

    printf("Enter a number: ");
    scanf("%d", &number);

    if(palindrome(number))
        printf("%d is a Palindrome number.\n", number);
    else
        printf("%d is Not a Palindrome number.\n", number);

    return 0;
}

```

Output:

Is a Palindrome number



Is not a Palindrome number



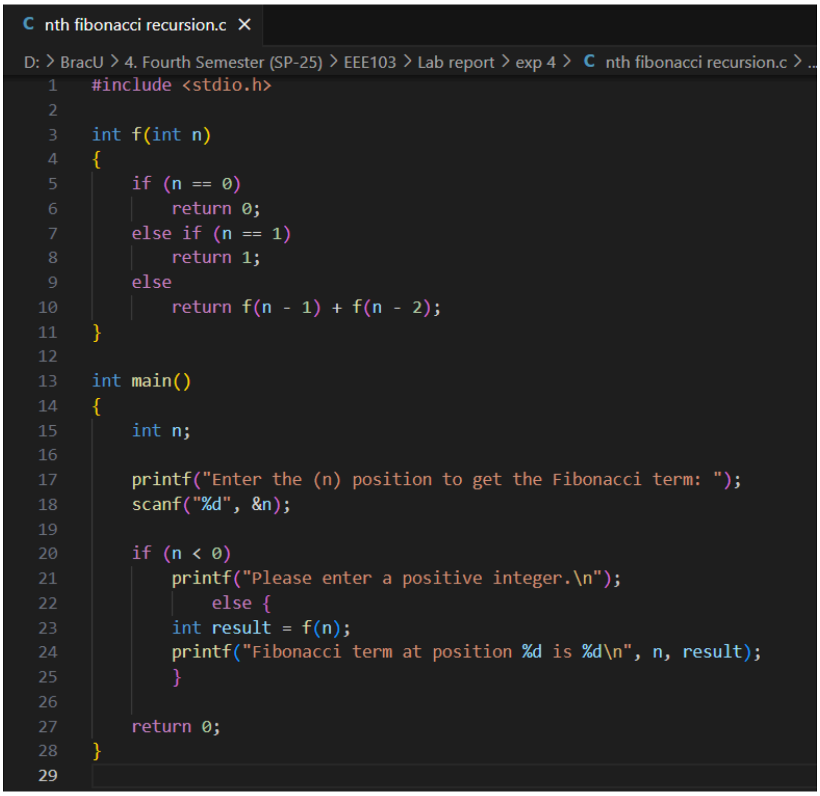
b. Write a C program to get the nth Fibonacci Term using recursion.

Code:

```
#include <stdio.h>

int f(int n)

{

if (n == 0)

return 0;

else if (n == 1)

return 1;

else

return f(n - 1) + f(n - 2);

}
```

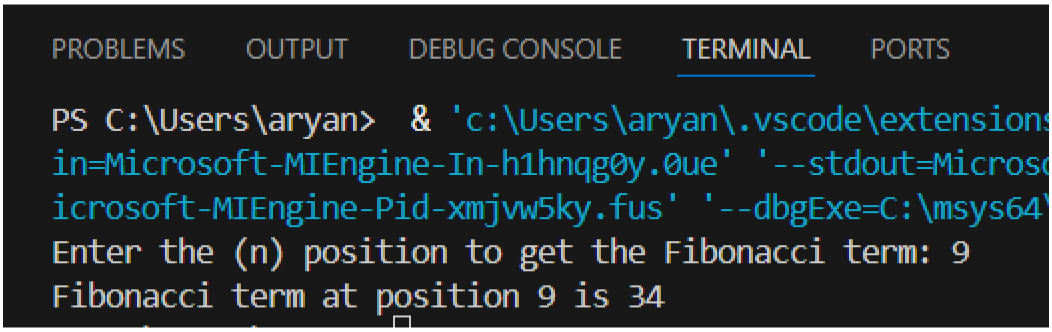
```
int main()

{

int n;

printf("Enter the (n) position to get the Fibonacci term: ");

scanf("%d", &n);

if (n < 0)

printf("Please enter a positive integer.\n");

else {

int result = f(n);

printf("Fibonacci term at position %d is %d\n", n, result);

}

return 0;

}
```

<u>Screenshot of code:</u>




```c
#include <stdio.h>

int f(int n)
{
    if (n == 0)
        return 0;
    else if (n == 1)
        return 1;
    else
        return f(n - 1) + f(n - 2);
}

int main()
{
    int n;

    printf("Enter the (n) position to get the Fibonacci term: ");
    scanf("%d", &n);

    if (n < 0)
        printf("Please enter a positive integer.\n");
            else {
        int result = f(n);
        printf("Fibonacci term at position %d is %d\n", n, result);
        }

    return 0;
}
```

Output:



c. Write a C program to find the sum of digits of a given number using recursion.

Code:

```
int main()

{

int n;

printf("Enter a number: ");

scanf("%d", &n);

if (n < 0)

printf("Please enter a positive number.\n");

else {

int result = sumdigits(n);

printf("Sum of digits of %d is %d\n", n, result);

}

return 0;

}
```

Screenshot of code:

C sum of d recursion.c ×

D: > BracU > 4. Fourth Semester (SP-25) > EEE103 > Lab report > exp 4 > C sum of d recursion.c > ...

```c
#include <stdio.h>

int sumdigits(int num)
{
    if (num == 0)
        return 0;
    else
        return (num % 10) + sumdigits(num / 10);
}

int main()
{
    int n;

    printf("Enter a number: ");
    scanf("%d", &n);

    if (n < 0)
        printf("Please enter a positive number.\n");
    else {
        int result = sumdigits(n);
        printf("Sum of digits of %d is %d\n", n, result);
    }

    return 0;
}
```

Output:

PROBLEMS OUTPUT DEBUG CONSOLE TE

```
PS C:\Users\aryan>  & 'c:\Users\arya
in=Microsoft-MIEngine-In-os0t55ib.p1
icrosoft-MIEngine-Pid-hmjkg5di.mrp'
Enter a number: 2468
Sum of digits of 2468 is 20
```